

KOTKA RANKKI, Finland

WMO: 029760

Lat: 60.375N

Lon: 26.959E

Elev: 13

StdP: 101.17

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-20.4	-17.4	-22.7	0.5	-20.3	-19.6	0.7	-17.3	14.7	-0.3	13.0	-1.3	3.9	10	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	24.6	20.1	23.1	19.3	21.7	18.3	21.1	23.5	19.9	22.6	18.8	21.2	3.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	14.9	22.8	18.8	13.7	21.7	17.7	12.7	20.5	61.1	23.4	57.0	22.7	53.3	21.3	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.8	11.3	10.1	-21.0	26.5	4.7	2.3	-24.4	28.2	-27.2	29.5	-29.8	30.8	-33.3	32.5		
(5)			-21.3	21.5	4.7	1.5	-24.6	22.6	-27.4	23.5	-30.0	24.3	-33.4	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-4.4	-5.4	-2.4	2.8	8.8	13.9	17.9	17.1	12.7	6.9	2.3	-1.1		
	DBStd	8.85	5.58	6.04	4.07	2.82	3.39	2.90	2.84	2.53	2.81	3.32	3.68	4.75		
	HDD10.0	2233	446	433	384	216	64	2	0	0	9	104	230	344		
	HDD18.3	4618	704	666	643	466	297	136	44	54	171	356	480	603		
	CDD10.0	707	0	0	0	1	26	119	243	222	89	8	0	0		
	CDD18.3	49	0	0	0	0	0	3	29	17	0	0	0	0		
	CDH23.3	104	0	0	0	0	0	6	74	24	0	0	0	0		
	CDH26.7	6	0	0	0	0	0	1	5	0	0	0	0	0		
	WSAvg	5.0	5.5	5.3	5.0	4.4	4.3	4.3	4.0	4.3	4.8	5.6	6.0	6.2		
	PrecAvg	579	44	42	34	30	35	55	50	60	53	58	62	45		
(15) Precipitation	PrecMax	706	83	99	63	64	84	129	102	128	104	112	133	81		
	PrecMin	414	7	2	5	7	7	6	13	3	12	18	7	10		
	PrecStd	76	18	27	14	17	23	31	28	33	27	23	29	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	3.2	5.6	13.1	21.2	23.9	27.4	25.5	19.8	14.2	9.6	7.1		
		MCWB	3.8	1.8	3.0	9.0	14.4	18.4	21.6	20.5	16.3	12.7	8.9	6.4		
	2%	DB	3.3	2.0	3.9	9.8	17.8	21.5	25.4	23.8	18.3	13.1	8.8	6.2		
		MCWB	2.8	1.5	2.0	6.5	13.2	17.1	20.5	19.6	15.6	11.9	8.2	5.7		
	5%	DB	2.4	1.5	2.7	8.1	15.7	19.9	23.9	22.5	17.3	12.2	8.0	5.3		
		MCWB	2.0	1.1	1.4	5.5	12.2	16.1	19.9	18.7	15.1	11.2	7.4	4.8		
	10%	DB	1.5	1.0	1.8	6.7	13.7	18.4	22.5	21.1	16.5	11.3	7.1	4.2		
		MCWB	1.1	0.7	1.0	4.6	10.6	15.2	19.2	17.9	14.6	10.3	6.6	3.7		
	0.4%	WB	4.0	2.2	3.6	9.1	15.6	19.8	23.2	21.9	17.3	13.2	9.2	6.7		
		MCDB	4.1	2.7	5.0	12.3	18.5	23.1	25.7	24.3	18.7	13.8	9.3	7.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	3.0	1.6	2.4	7.2	14.0	18.1	21.8	20.5	16.4	12.1	8.2	5.8		
		MCDB	3.2	1.9	3.3	9.3	16.8	20.7	24.1	23.1	17.6	12.9	8.6	6.1		
	5%	WB	2.1	1.2	1.7	5.9	12.5	16.8	20.6	19.4	15.7	11.4	7.5	4.8		
		MCDB	2.3	1.4	2.4	7.6	15.3	19.1	23.1	21.8	16.8	12.1	7.9	5.1		
	10%	WB	1.2	0.7	1.2	4.9	11.0	15.6	19.5	18.3	15.0	10.5	6.6	3.8		
		MCDB	1.5	1.0	1.7	6.4	13.5	17.9	22.1	20.5	16.2	11.1	7.0	4.2		
	5% DB	MDBR	3.9	4.4	5.0	4.8	5.5	5.2	5.2	4.8	4.0	3.0	2.9	3.3		
		MCDBR	2.7	2.9	4.5	6.7	7.9	7.2	6.7	6.0	4.7	3.0	2.9	2.5		
		MCWBR	2.8	2.9	3.3	4.1	4.2	3.7	3.3	3.1	3.1	2.8	3.0	2.6		
		MCWBR	2.7	2.8	3.8	6.0	7.1	6.2	5.8	5.1	3.9	2.9	2.7	2.4		
(35) Mean Daily Temperature Range	5% WB	MCWBR	2.9	2.8	2.9	3.9	4.2	3.8	3.3	3.1	2.9	2.7	2.8	2.6		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.311	0.333	0.371	0.369	0.365	0.387	0.380	0.353	0.334	0.305	0.382		
	taud		2.139	2.167	2.215	2.285	2.404	2.458	2.417	2.433	2.489	2.388	2.207	2.596		
	Ebn at Noon		466	665	783	823	857	866	834	810	772	649	430	326		
	Edh at Noon		42	73	96	109	104	100	103	94	75	60	40	29		
(44) All-Sky Solar Radiation	RadAvg		0.25	0.86	2.35	3.87	5.30	5.52	5.52	4.21	2.50	1.03	0.28	0.13		
	RadStd		0.06	0.18	0.29	0.42	0.57	0.37	0.49	0.43	0.29	0.15	0.05	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	-166	-176	N/A	N/A

Nomenclature: See separate page